End of Year Maths Assessment

Year 4

This resource corresponds with the New National Curriculum objectives for Year 4, and has been designed to be used as an aide in assessing pupils' end of year mathematical knowledge at the end of Year 4.

This assessment can be used to assist teachers with end of year ability levelling, planning/assessment and as an ideal support tool for parents' evenings/progress meetings etc.

The content addresses each curricular objective outlined in the New National Curriculum in chronological order to give insight into strengths or gaps in each child's mathematical knowledge.

More Assessment resources.

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End of Year Maths Assessment – Year 4

Year 4 National Curriculum Objectives:

Number – number and place value

1. (4N1) Count in multiples of 6, 7, 9, 25 and 1000

5. (4N2a) Order and compare numbers beyond 1,000

2. (4N2b) Find 1,000 more or less than a given number

4. (4N3a) Recognise the place value of each digit in a four-digit number (thousands, hundreds, tens, and ones)

8. (4N3b) Read Roman numerals to 100 (I to C) and know that over time, the numeral system changed to include the concept of zero and place value

6. (4N4a) Identify, represent and estimate numbers using different representations

7. (4N4b) Round any number to the nearest 10, 100 or 1,000

3. (4N5) Count backwards through zero to include negative numbers

1, 2, 3, 4, 5, 6, 7 & 8. (4N6) Solve number and practical problems that involve 4N1 - 4N5 and with increasingly large positive numbers

Number – addition and subtraction

9, 10, 11 & 12. (4C2) Add and subtract numbers with up to 4 digits using the formal written methods of columnar addition and subtraction where appropriate

13, 14, 15 & 16. (4C3) Estimate and use inverse operations to check answers to a calculation

17. (4C4) Solve addition and subtraction two-step problems in contexts, deciding which operations and methods to use and why

Number – multiplication and division

18, 19 & 20. (4C6a) Recall multiplication and division facts for multiplication tables up to 12 × 12

21, 22 & 23. (4C6b) Use place value, known and derived facts to multiply and divide mentally, including: multiplying by 0 and 1; dividing by 1; multiplying together three numbers

24. (4C6c) Recognise and use factor pairs and commutativity in mental calculations

25 & 26. (4C7) Multiply two-digit and three-digit numbers by a one-digit number using formal written layout

27. (4C8) Solve problems involving multiplying and adding, including using the distributive law to multiply two digit numbers by one digit, integer scaling problems and harder correspondence problems such as *n* objects are connected to *m* objects

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End of Year Maths Assessment – Year 4

Year 4 National Curriculum Objectives:

Number – fractions (including decimals)

29. (4F1) Count up and down in hundredths; recognise that hundredths arise when dividing an object by one hundred and dividing tenths by ten

28. (4F2) Recognise and show, using diagrams, families of common equivalent fractions

31. (4F4) Add and subtract fractions with the same denominator

33. (4F6a) Recognise and write decimal equivalents to 1/4 , 1/2 , 3/4

33. (4F6b) Recognise and write decimal equivalents of any number of tenths or hundredths

34. (4F7) Round decimals with one decimal place to the nearest whole number

35. (4F8) Compare numbers with the same number of decimal places up to two decimal places

32. (4F9) Find the effect of dividing a one- or two-digit number by 10 and 100, identifying the value of the digits in the answer as ones, tenths and hundredths

30. (4F10a) Solve problems involving increasingly harder fractions to calculate quantities, and fractions to divide quantities, including non-unit fractions where the answer is a whole number

36 & 37. (4F10b) Solve simple measure and money problems involving fractions and decimals to two decimal places

Measurement

41. (4M1) Compare different measures, including money in pounds and pence

* (4M2) Estimate different measures, including money in pounds and pence

42. (4M4a) Read, write and convert time between analogue and digital 12 clocks

42. (4M4b) Read, write and convert time between analogue and digital 24-hour clocks

43. (4M4c) Solve problems involving converting from hours to minutes; minutes to seconds; years to months; weeks to days

38. (4M5) Convert between different units of measure [for example, kilometre to metre; hour to minute]

39. (4M7a) Measure and calculate the perimeter of a rectilinear figure (including squares) in centimetres and metres

40. (4M7b) Find the area of rectilinear shapes by counting squares

17, 27, 36 & 37. (4M9) Calculate different measures, including money in pounds and pence

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End of Year Maths Assessment – Year 4

Year 4 National Curriculum Objectives:

Geometry – properties of shapes

44. (4G2a) Compare and classify geometric shapes, including quadrilaterals and triangles, based on their properties and sizes

46. (4G2b) Identify lines of symmetry in 2-D shapes presented in different orientations

47. (4G2c) Complete a simple symmetric figure with respect to a specific line of symmetry

45. (4G4) Identify acute and obtuse angles and compare and order angles up to two right angles by size

Geometry – position and direction

49. (4P2) Describe movements between positions as translations of a given unit to the left/right and up/down

48. (4P3a) Describe positions on a 2-D grid as coordinates in the first quadrant

50. (4P3b) Plot specified points and draw sides to complete a given polygon

Statistics

51. (4S1) Interpret and present discrete and continuous data using appropriate graphical methods, including bar charts and time graphs

52. (4S2) Solve comparison, sum and difference problems using information presented in bar charts, pictograms, tables and other graphs

*these objectives are practical so are not included in this assessment

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End of Year Maths Assessment – Year 4

Number and Place Value

1. Complete the next FOUR multiples of 6, 7, 9, 25 and 1,000.

18	24	30	_____	_____	_____	_____
14	21	28	_____	_____	_____	_____
0	9	18	_____	_____	_____	_____
75	100	125	_____	_____	_____	_____
3,000	4,000	5,000	_____	_____	_____	_____

2. Find 1,000 more or less than the given number.

1,000 less	number	1,000 more
	3,983	
	6,003	
	2,903	
	1,019	
	9,192	

3. Complete the sequence of numbers.

4 3 2 1 [ ] [ ] [ ] [ ] [ ] [ ]

4. Write the value of the underlined digit.

648 _____

1493 _____

6030 _____

2997 _____

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5. Use the symbols > or < to compare these numbers.

3,683 ☐ 3,468 7,283 ☐ 8,723 6,104 ☐ 6,201

6. Write the following numbers in words.

483 _____

2,940 _____

5,701 _____

7. Round the following numbers to the nearest 10.

25 _____ 84 _____ 36 _____

Round the following numbers to the nearest 100.

770 _____ 342 _____ 181 _____

Round the following numbers to the nearest 1,000.

7,293 _____ 1,920 _____ 5,299 _____

8. Write the number that these Roman numerals represent.

XC _____ XL _____

XVI _____ XIX _____

LXI _____ C _____

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Addition and Subtraction

9. 126 + 502 =

10. 2,637 + 5,443 =

11. 927 – 486 =

12. 7,028 – 4,529 =

13. 372 + ☐ = 736

14. ☐ + 582 = 724

15. ☐ – 275 = 883

16. 638 – ☐ = 144

17. Miles buys a bottle of juice for £1.50, a banana for 47p and a granola bar for 83p. He pays with a £5 note. How much change should Miles be given?

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Multiplication and Division

18. 8 x 9 = ☐

19. 12 x 6 = ☐

20. 63 ÷ 7 = ☐

21. 8 ÷ 1 = ☐

22. 7 x 0 = ☐

23. 3 x 6 x 2 = ☐

24. Identify the factor pairs of these numbers.

24 = _____

32 = _____

25. 48 x 9 =

26. 726 x 8 =

27. Rosa buys 4 movie tickets at £16 each. At the till, she also uses a voucher that gives her £5 off the final bill. How much does Rosa pay in total?

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Fractions (including decimals)

28. Circle the equivalent fraction.

$\frac{1}{2}$ is equivalent to: $\frac{5}{6}$ $\frac{6}{12}$ $\frac{8}{9}$

$\frac{1}{4}$ is equivalent to: $\frac{3}{8}$ $\frac{4}{6}$ $\frac{3}{12}$

29. Complete the sequences.

$\frac{41}{100}$	$\frac{42}{100}$	$\frac{43}{100}$	$\frac{44}{100}$				
0.26	0.27	0.28	0.29				

30. Calculate these fractions of amounts.

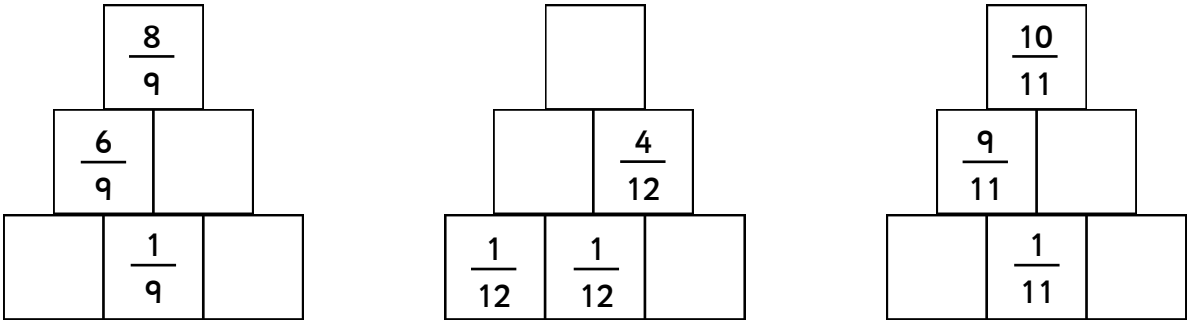
$\frac{1}{4}$ of £36 = ☐

$\frac{1}{7}$ of £63 = ☐

$\frac{2}{3}$ of £18 = ☐

$\frac{3}{4}$ of £36 = ☐

31. Use addition and subtraction to complete the missing fraction pyramids.



32. Divide the following numbers by 10 and 100.

26 ÷ 10 = ☐

67 ÷ 10 = ☐

74 ÷ 100 = ☐

82 ÷ 100 = ☐

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33. Draw a line to match each fraction to its decimal equivalent.

$\frac{7}{14}$	0.2
$\frac{2}{100}$	0.75
$\frac{3}{12}$	0.02
$\frac{12}{16}$	0.5
$\frac{1}{5}$	0.25

34. Round the following decimals to the nearest whole number.

7.4 ☐ 4.2 ☐ 6.8 ☐ 22.6 ☐

35. Order these decimals from smallest to largest.

2.25	2.05	0.52	2.20	2.5
☐	☐	☐	☐	☐

36. Ross buys 6 bags of flour. Each bag weighs 1.56kg. What is the total weight of the 6 bags of flour?

37. Haruki buys 8 bars of chocolate to share amongst his friends. Each bar cost £2.49. How much did the 8 bars cost altogether?

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Measurement

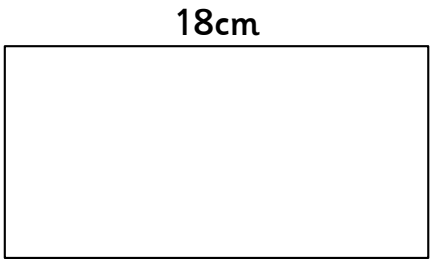
38. Convert the following hours to minutes.

3hrs [mins] 6hrs [mins] 2½hrs [mins]

Convert the following kilometres to metres.

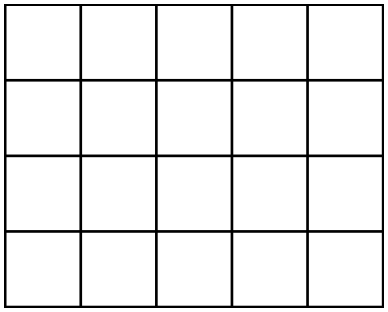
4km [m] 5km [m] 3½km [m]

39. Calculate the perimeter of the following shape.



Perimeter = _____ cm

40. Each square is 2cm long and 2cm tall. Calculate the area of the shape.



Area = _____ cm^2

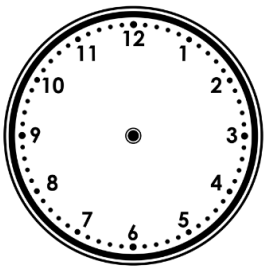
41. Order the following amounts from largest to smallest.

£2.30 59p £1.98 240p £1.75

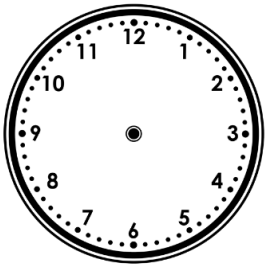
[ ] [ ] [ ] [ ] [ ]

42. Show the given 12 hr and 24hr times on an analogue clock.

4:35pm



23:23



43. It's Frank's birthday in exactly 3 weeks. How many days is that? _____

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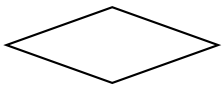
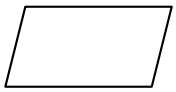
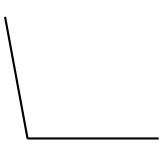
Geometry (properties of shapes)

44. Draw a line to connect each shape to the correct name and properties.

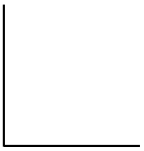
rhombus		1 pair of parallel sides
trapezium		4 equal sides and opposite angles are the same size
isosceles triangle		a triangle with two equal sides
parallelogram		opposite sides are parallel and equal in length

45. Add the correct label to the following angles.

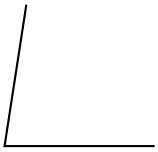
acute	obtuse	right angle



_____

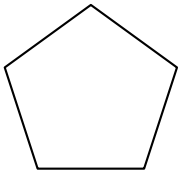
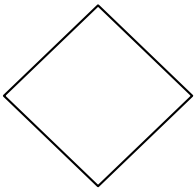


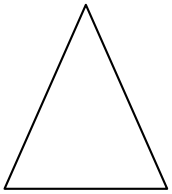
_____



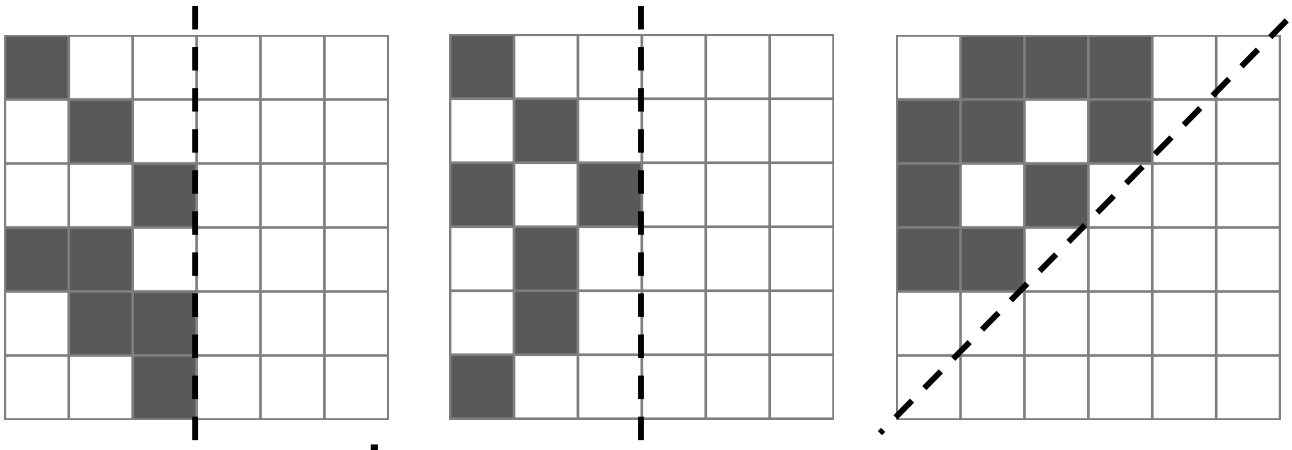
_____

46. Draw lines of symmetry in the following 2D shapes.



47. Shade in the squares to complete these symmetrical patterns.



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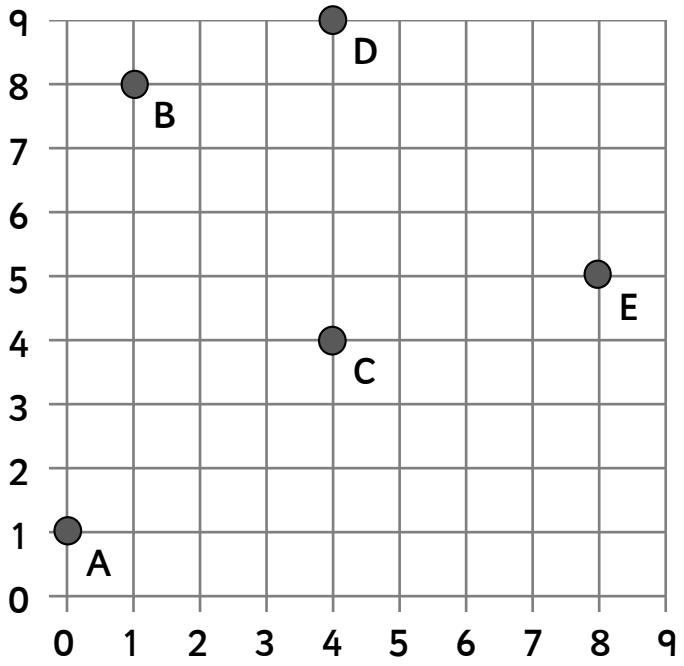
48. Write down the coordinates for each letter.



A = (,)

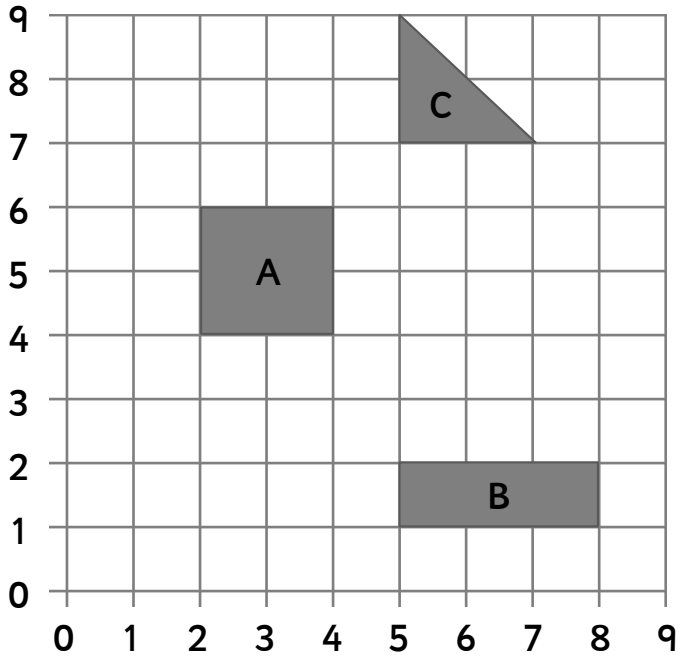
B = (,)

C = (,)

D = (,)

E = (,)

49. Follow the instructions and draw each shape in its new position.



- Translate shape A 2 squares to the left and 4 squares down.
- Translate shape B 3 squares to the left and 6 squares up.
- Translate shape C 5 squares down and 1 square to the right.

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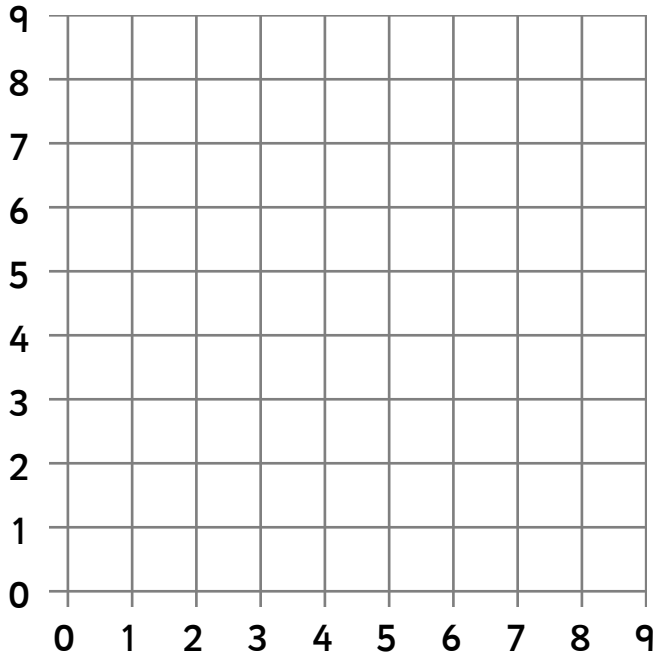
50. Plot the following coordinates on the grid to create a shape.



(1,6)

(2,2)

(6,2)

(7,6)

(4,8)

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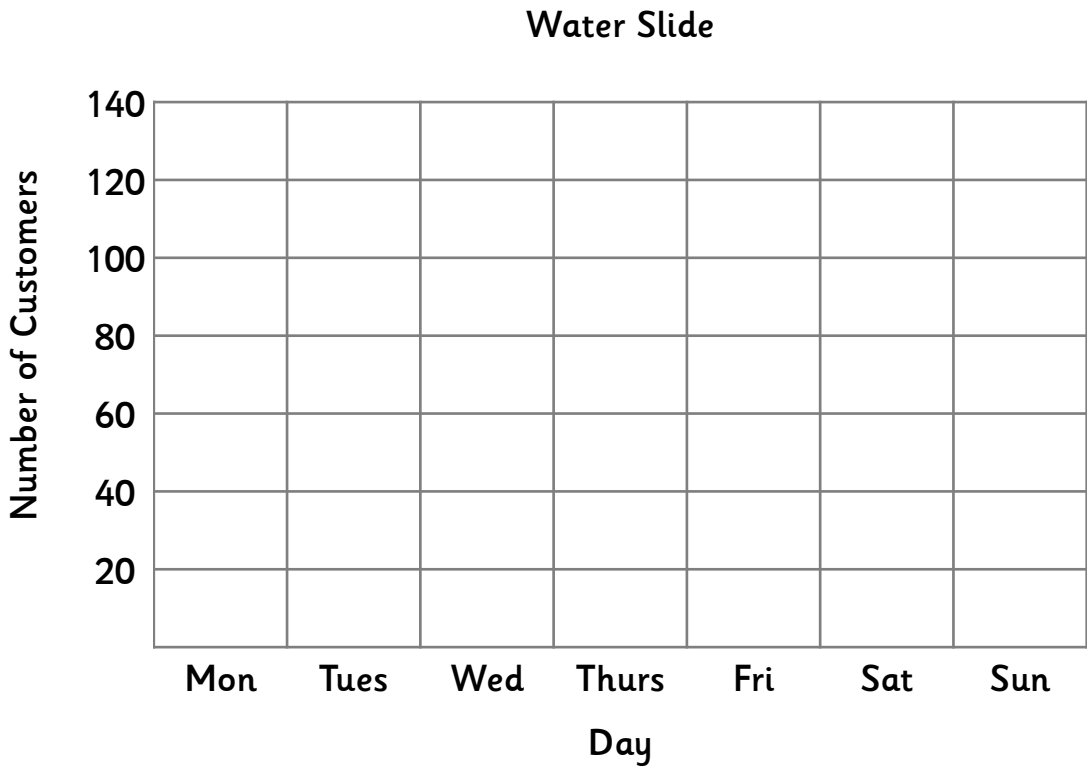
51. Complete the graph using the data below.

	Mon	Tues	Wed	Thurs	Fri	Sat	Sun
Number of customers at the water slide	0	40	20	60	120	140	130



52. What day is the water slide the busiest? _____

What day is the water slide most likely closed? _____

How many more customers came on Friday than Tuesday? _____

How many customers used the water slide in total this week? _____

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End of Year Maths Assessment – Year 4

Number and Place Value

1. 36; 42; 48; 54
   35; 42; 49; 56
   27; 36; 45; 54
   150; 175; 200; 225
   6,000; 7,000; 8,000; 9,000

2.

1000 less	given number	1000 more
2,983	3,983	4,983
5,003	6,003	7,003
1,903	2,903	3,903
19	1,019	2,019
8,192	9,192	10,192

3. 0; -1; -2; -3; -4; -5

4. 8 ones
   9 tens
   0 hundreds
   2 thousands

5. >, <, <

6. four hundred and eighty-three
   two thousand, nine hundred and forty
   five thousand, seven hundred and one

7. 30; 80; 40
   800; 300; 200
   7,000; 2,000; 5,000

8.

XC	90	XL	40
XVI	16	XIX	19
LX1	61	C	100

Addition and Subtraction

9. 628 10. 8,080 11. 441 12. 2,499

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13. 364 14. 142 15. 1,158 16. 494

17. £2.20

Multiplication and Division

18. 72 19. 72 20. 9 21. 8 22. 0 23. 36

24. 1 x 24; 2 x 12; 3 x 8; 4 x 6
1 x 32; 2 x 16; 4 x 8

25. 432

26. 5,808

27. £59

Fractions (including decimals)

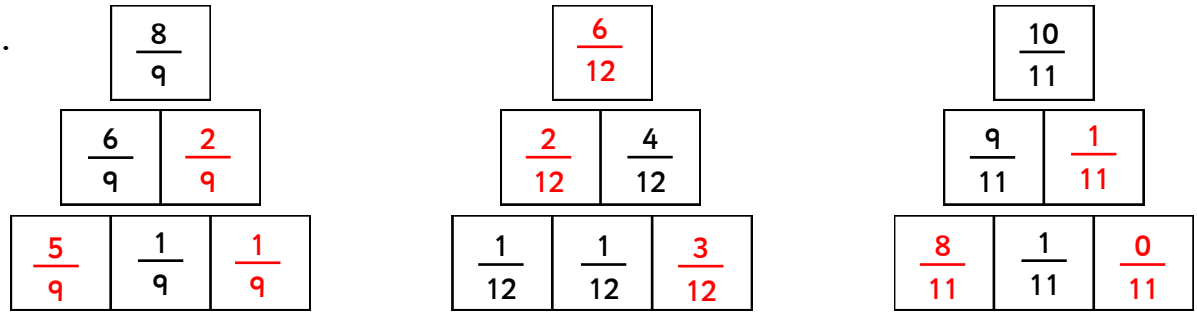
28. $\frac{1}{2}$ is equivalent to: $\frac{5}{6}$ $\frac{6}{12}$ $\frac{8}{9}$ $\frac{1}{4}$ is equivalent to: $\frac{3}{8}$ $\frac{4}{6}$ $\frac{3}{12}$

29. $\frac{45}{100}$ $\frac{46}{100}$ $\frac{47}{100}$ $\frac{48}{100}$
0.3 0.31 0.32 0.33

30. $\frac{1}{4}$ of £36 = £9 $\frac{1}{7}$ of £63 = £9

$\frac{2}{3}$ of £18 = £12 $\frac{3}{4}$ of £36 = £27

31.



32. 26 ÷ 10 = 2.6 67 ÷ 10 = 6.7

74 ÷ 100 = 0.74 82 ÷ 100 = 0.82

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33.

$\frac{7}{14}$	0.2
$\frac{2}{100}$	0.75
$\frac{3}{12}$	0.02
$\frac{12}{16}$	0.5
$\frac{1}{5}$	0.25

34. 7; 4; 7; 23

35. 0.52; 2.05; 2.20; 2.25; 2.5

36. 9.36kg

37. £19.92

Measurement

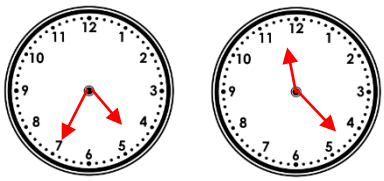
38. 180mins; 360mins; 150mins 4,000m; 5,000m; 3,500m

39. 50cm

40. 80cm²

41. 240p; £2.30; £1.98; £1.75; 59p

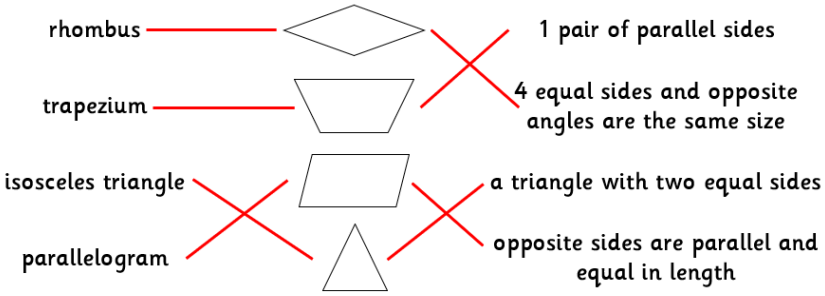
42.



43. 21 days

Geometry (properties of shapes)

44.



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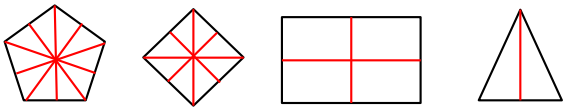
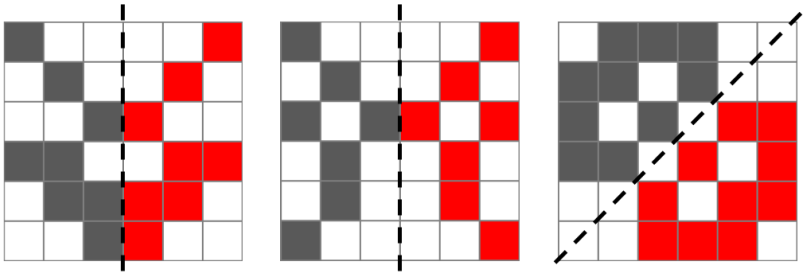
45. Obtuse angle; right angle; acute angle

46.



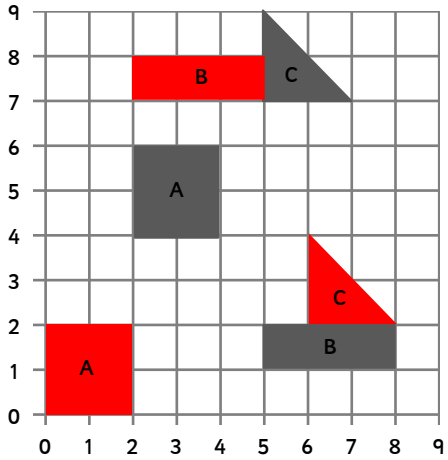
47.



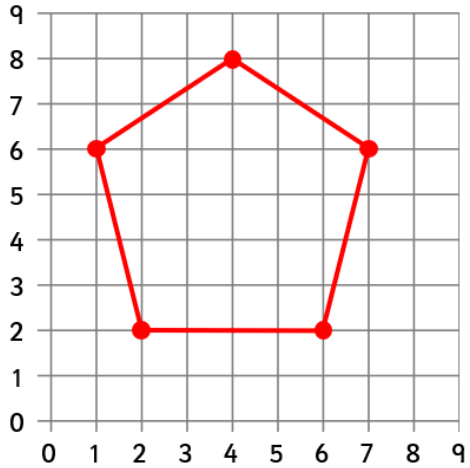
Geometry (position and direction)

48. A = (0,1), B = (1,8), C = (4,4), D = (4,9), E = (8,5)

49.

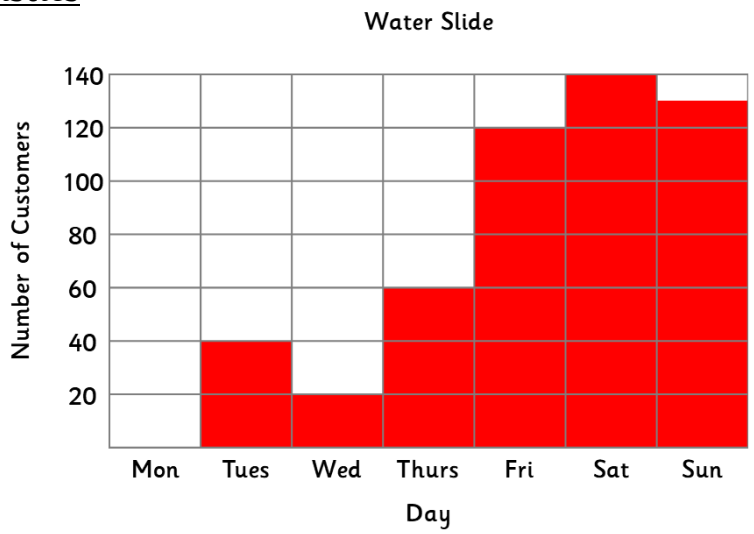


50.



Statistics

51.



52. Saturday; Monday; 80; 510

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